Anderson Junior College
2009 JC 1 Promotional Examination
H2 Chemistry Solutions

<u>Section A – Free Response</u>

1 (a) (i) To prevent reactants NH_3 and/or HCl from reacting with steel as both are corrosive gases.

(ii)

$$
\begin{tikzcd}
NH_3(g) & + & HCl(g) \arrow[r, "-176"] & NH_4Cl(s) \\
 & \tfrac{1}{2}N_2(g) + 2H_2(g) + \tfrac{1}{2}Cl_2(g) \arrow[ul, "-45.9"'] \arrow[ur, "-92.3"] \arrow[dl, "+634"'] \arrow[dr, "-243"] & & \\
NH_4^{+}(g) & + & Cl^{-}(g) \arrow[uur, "LE"'] &
\end{tikzcd}
$$

By Hess Law,
LE = -634 + 243 + (-45.9) + (-92.3) + (-176) = - 705 kJ mol^{-1}

(iii) LE (NH_4Br) will be numerically smaller than LE (NH_4Cl).
- Both Br^- and Cl^- are singly charged ions
- Ionic radius of Br^- bigger than that of Cl^-
- weaker attraction between NH_4^+ and Br^- than between NH_4^+ and Cl^- ions.

(b) (i)
$$
\begin{aligned}
\Delta G^{\ominus} &= \Delta H^{\ominus} - T\Delta S^{\ominus} \\
&= (-176) - (298)(-0.284) \\
&= -91.4 \text{ kJ mol}^{-1} < 0
\end{aligned}
$$

(ii) Since both $\Delta H^{\ominus}$ and $\Delta S^{\ominus}$ are both negative, $\Delta G^{\ominus}$ is negative only under low temp. Reaction spontaneous only under low temperature.

(iii) For $\Delta G^{\ominus} = 0$

$$T = \frac{\Delta H^{\ominus}}{\Delta S^{\ominus}}$$

$$T = \frac{-176}{-0.284} = 619.7 \text{ K}$$

(iv) The increase in entropy for gaseous particles will be much more than the increase in entropy for a solid when the temperature increases from 298 K to 620 K.

The $\Delta S^{\ominus}$ at 620 K is more negative / numerically larger than $\Delta S^{\ominus}$ at 298 K.
As a result, T should be lower than calculated.

(c) **(i)** $NH_4Cl\ (s)\ (+\ aq) \rightarrow NH_4^+(aq) + Cl^-\ (aq)$

(ii) $n(NH_4Cl) = 5 / 53.5 = 0.0935$ mol
Heat absorbed = 15 x 0.0935 = 1.40 kJ = 1400 J = $mc\theta$
$1400 = 125 \times 4.2 \times \theta$
$\theta = 2.67$ ºC
Final temperature = 20 – 2.67 = 17.3 ºC

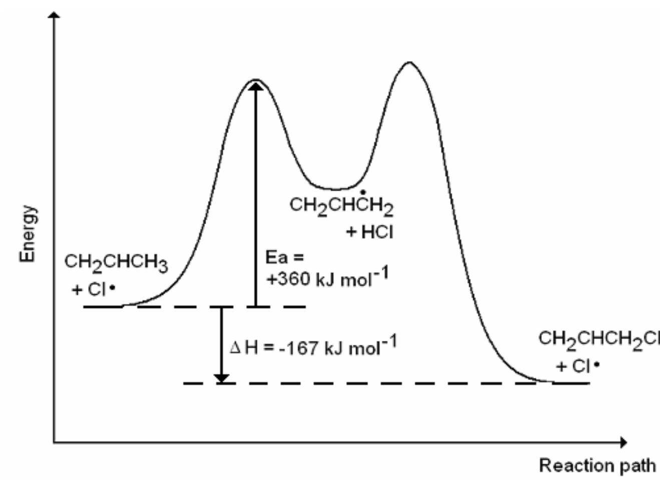
(d) $n_{NH_3} = \dfrac{10.00}{14.0+3(1.0)} = 0.5882\,\text{mol}$ $\qquad n_{HCl} = \dfrac{10.00}{1.0+35.5} = 0.2740\,\text{mol}$

HCl is the limiting reagent.

Amt. of excess NH_3 = 0.5882 – 0.2740 = 0.3142 mol

Final pressure of system = $\dfrac{(0.3142)(8.31)(273+25)}{(5+5)\times 10^{-3}} = 7.78\times 10^4\,\text{Pa (3 s.f.)}$

2 **(a)** **(i)**



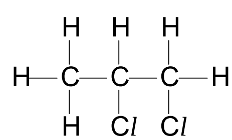
(ii) The catalyst $Cl\bullet$ is in the same phase as the reactant propene.

(iii) Ea of the alternative pathway in the presence of a catalyst is lower so more molecules to possess energy ≥ lowered Ea, and so frequency of effective collision increases and thus faster rate of reaction.

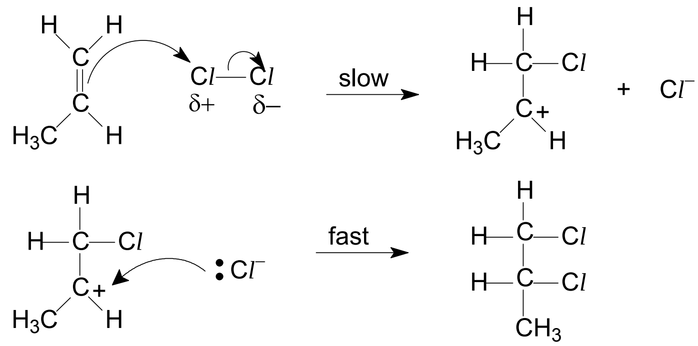
(iv) Rate increases despite all concentrations of reactants remaining the same, thus the value of k must also increase.

(b) **(i)** Empirical formula of compound A is $C_3H_6Cl_2$.

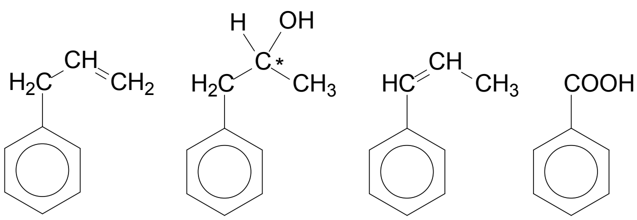
Structural formula: $CH_3CHClCH_2Cl$ or



(ii) Type of reaction: electrophilic addition



(c) 1. B undergoes electrophilic addition with $Br_2 \Rightarrow$ B is an alkene.
2. B undergoes (strong) oxidation to give CO_2 gas $\Rightarrow$ B contains a terminal double bond.
3. B reacts with H_2O in an (electrophilic) addition reaction to give C.
4. C undergoes elimination of H_2O / dehydration to give an alkene D.
5. E undergoes acid-base / neutralisation with NaOH(aq $\Rightarrow$ E contains COOH / acidic



B C D E

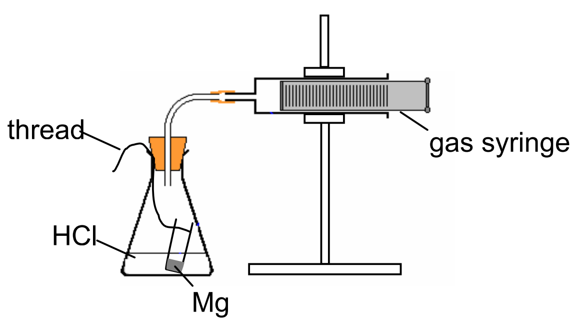
(d) More energy needed to break the stronger dipole–dipole attractions between C_3H_5Cl molecules than the weaker van der Waals forces between C_3H_6 molecules.
Hence, 3–chloropropene has higher boiling point than propene.

Section B – Structured Questions

1 **(a)** For a reaction aA + bB → products

- Rate equation is rate = $k[A]^x[B]^y$
- x and y are the orders of reaction with respect to A and B respectively.
- k is the rate constant

(b) (i)



(ii)
- Place excess HC*l* into the flask and the Mg strip in the small tube.
- Start the reaction and start the stopwatch simultaneously.
- Measure the volume of gas at regular time intervals until the reaction is complete.

(iii) Plot a graph of V_{H2} against time or graph of ($V_{final} - V_{H2}$) against time and check that the half-life is constant.

(c) (i) Comparing experiments 1 & 3, when [HCl] doubles, rate doubles.
$\Rightarrow$ reaction is first order with respect to HCl

$$\frac{rate_2}{rate_1} = \frac{0.054}{0.024} = \frac{0.15}{0.10}\left(\frac{0.15}{0.10}\right)^x \Rightarrow 1.5 = 1.5^x \Rightarrow x = 1$$

$\Rightarrow$ reaction is first order with respect to sucrose

rate = k[HC*l*][sucrose]

(ii) Using experiment 1, 0.024 = k(0.10)(0.10)
$\Rightarrow k = 2.4\ mol^{-1}\ dm^3\ s^{-1}$

(iii) 1.5 s since [HC*l*] is doubled that of experiment 1

(iv) I: first order

II: At high [sucrose], all the enzyme (sucrase) has been converted into enzyme-substrate complex.

2 **(a)** (i) $H_3BO_3 + NH_3 \rightarrow BN + 3\ H_2O$

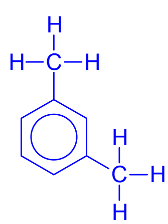
(ii) $\angle OBO = 120^o$
$\angle BOH = 105^o$ or 104.5^o

(b) Both hexagonal-BN and cubic-BN are giant molecules / macromolecular / giant covalent

h-BN
- (strong) B-N covalent bond btw atoms within layers but
- Weak Van der Waals forces between layers
- allowing easy sliding of layers over each other

c-BN
- strong covalent bonds between B and N atoms thus
- rigid structure/extensive network

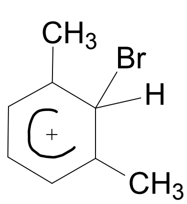
3 **(a)** (i)



(ii) $FeBr_3$ or $AlBr_3$

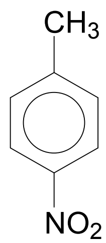
(iii) Electrophilic substitution

(iv)



Type(s) of hybridization: sp^2 and sp^3

(b)



I conc. HNO_3 and conc. H_2SO_4

II $KMnO_4$, dilute H_2SO_4

Section C - MCQ

1	**C**	**6**	**A**	**11**	**D**	**16**	**B**	**21**	**B**	**26**	**D**
2	**C**	**7**	**A**	**12**	**B**	**17**	**B**	**22**	**C**	**27**	**D**
3	**B**	**8**	**D**	**13**	**B**	**18**	**A**	**23**	**D**	**28**	**C**
4	**B**	**9**	**C**	**14**	**C**	**19**	**D**	**24**	**C**	**29**	**A**
5	**B**	**10**	**A**	**15**	**A**	**20**	**B**	**25**	**D**	**30**	**C**